IBVAP Command Center

Frontend guide: kya hai, kab kya hota hai, aur kaise chalta hai · Smart India Hackathon, PS 26187

> **Ek line mein:** IBVAP border ke purane CCTV cameras ko AI se "smart" banata hai, bina naye hardware ke. Yeh frontend woh **dashboard** hai jo control room mein operator dekhta hai: live cameras, alerts, map, reports.

1. Chalana kaise hai

1. `frontend` folder mein terminal kholo aur `npm run dev` chalao.
2. Browser mein `http://localhost:5173` kholo.
3. Role chuno (Operator / Commander / Admin) aur **Enter Command Center** dabao. Password nahi hai.

2. Andar se kaise kaam karta hai

```
Page  →  Store (data yahan rehta hai)  →  Service  →  Mock (nakli)  YA  Real backend
```

- **Service:** `.env` mein `VITE_USE_MOCK=true` ho toh nakli backend, `false` ho toh asli FastAPI. Pages ko farq pata nahi chalta.
- **Mock:** browser ki memory mein 8 cameras aur 14 din ke alerts banata hai. Page reload karte hi sab reset.
- **Store:** login ke baad ek baar live alert stream se judta hai; toast, sound aur popup wahi chalata hai.

Mock mein kab	Apne aap kya hota hai
Har 5-10 second	Ek naya random alert aata hai
Har 4-9 second	Ek naya number plate read hota hai
Lagataar	Camera feeds par detection boxes hilte rehte hain
Lagbhag 30% baar	Pichla alert dobara aata hai (same object), jisse "incident" banta hai

3. Ek alert aane par kya-kya hota hai

1. **Alert aata hai.** System dekhta hai: yeh naya mamla hai ya kisi purane **incident** ka hissa?
   Same camera + same track ID (10 min ke andar), ya same camera + same type (2 min ke andar) = wahi incident.
2. **Naya incident:** camera feed par laal flash. CRITICAL / HIGH ho toh toast + beep + bell par laal pulse. Fence Intrusion ya Face Match ho toh poori screen par **CRITICAL ALERT** dialog.
3. **Purane incident ka repeat:** koi naya popup ya sound nahi, bas card par counter badhta hai (x2, x3).
4. **Koi dhyan na de:** CRITICAL alert har 30 second mein dobara yaad dilata hai.
5. **Operator Acknowledge dabata hai:** poora incident acknowledge hota hai aur **response time** record hota hai.
6. **Aage:** Escalate → Resolve (sirf Commander / Admin) ya False alarm. Har kadam user aur time ke saath history mein.

> **Backend link toot jaaye toh:** upar banner aata hai aur purana data dikhta rehta hai. Acknowledge ya notes karoge toh woh queue mein chale jaate hain ("2 pending sync") aur link wapas aate hi apne aap bhej diye jaate hain.

4. Kaun kya kar sakta hai

Kaam	Operator	Commander / Admin
Acknowledge, Escalate, False alarm, playbook, QRT notify	**Haan**	**Haan**
Alert Resolve karna	**Nahi**	**Haan**
Zones, watchlist, blacklist, settings badalna	**Sirf dekhna**	**Haan**
Audit Log page	**Nahi dikhta**	**Haan**

5. Har page kya karta hai

Command Center (home)
- 6 bade numbers, har ek ke saath 24 ghante ka graph
- 4 live cameras + daayein live alert feed
- Neeche pichhle 60 minute ki timeline

Live Cameras
- Layout 1x1 se 4x4; boxes / zones / track ID on-off
- Offline camera par "SIGNAL LOST" + last-seen time
- Tile par click: bada view + Snapshot button

Map View
- Camera markers, border ki dashed line (andaaze se)
- Naye alert par marker pulse karta hai + popup
- Filters, heat circles, bina internet ke grid view

Alerts
- Do tabs: Incidents (default) aur All alerts
- Filters, search, CSV export
- Row par click: Alert Detail khulta hai (neeche dekho)

ANPR (number plate)
- Live plate table; plate par click: gaadi ka route
- Kam confidence wala read operator theek kar sakta hai
- Blacklist: match hote hi alert banta hai

Face Watchlist
- Watchlist ke log, photos ke saath
- Matches: captured vs watchlist photo, similarity %
- Insaan Confirm / Reject karta hai (AI sirf madad)

Zones (virtual fence)
- Camera par polygon zone ya tripwire line banao
- Severity, kis par trigger, dwell time, schedule
- Saved zones har feed par dikhte hain; change log

Analytics aur Reports
- Graphs: time, camera, type, din vs raat, heatmap
- Average response time aur false alarm rate
- Shift report: CSV ya Print / PDF

Audit Log aur System Health
- Audit: kisne kya dekha, badla, export kiya
- Health: har outpost ke AI box ka CPU, disk, garmi, FPS
- Gadbad par warning; har 5 second refresh

Settings aur About
- Cameras, detection, notifications, C2 integration
- Privacy, security (auto-logout), users, Demo Mode
- About: architecture + 12 requirements ki table

Alert Detail page par kya milta hai

Hissa	Kya karta hai
Snapshot	Photo par object ka box aur zone highlight
Why it fired	Kitne frames par confirm hua, object class, confidence, visibility (fog / dust / low light)
Response playbook	Us alert type ke SOP steps; tick karo toh history mein save hota hai
Notify QRT	Quick Reaction Team ke liye SMS / WhatsApp message ka preview aur Send
Evidence	SHA-256 hash, "Integrity verified" badge, aur ZIP export (photo + hashes + kisne export kiya)
Workflow	Status buttons, live timer, history timeline, operator notes, map pin

6. Shortcuts aur Demo Mode

Key	Kaam
`A`	Saamne wala alert acknowledge (popup wala, ya jis card par mouse hai)
`M`	Alert sound mute / unmute
`1` `2` `3` `4`	Live Cameras ka layout
`?`	Shortcuts ki list
`Ctrl+Shift+D`	Demo Mode start / stop

> **Demo Mode (judges ke liye, lagbhag 2 minute):** random alerts ruk jaate hain aur ek scripted kahani chalti hai, neeche caption ke saath:
> patrol → gaadi detect → ANPR blacklist match → raat ka camera → fence intrusion (CRITICAL dialog) → acknowledge → wahi intruder do baar aur (ek hi incident, x3) → playbook ticks → QRT notify → evidence ZIP download → map par marker pulse → escalate → shift report → About page.

7. Abhi kya asli nahi hai

- Saara data nakli (mock) hai. Asli AI engine aur FastAPI backend abhi bane nahi hain.
- Video ki jagah animated placeholder hai; detection boxes random chalte hain.
- QRT message sach mein kahin nahi jaata. Face blur aur data retention sirf settings mein save hote hain.
- Real backend se judne wala code likha hai, par abhi tak kisi asli server ke saath chalaya nahi gaya.
- Map par border line andaaze se banayi hai, official nahi hai.
- Backend banane ke liye poora contract `frontend/docs/API_CONTRACT.md` mein hai.